

main

November 25, 2025

```
[1]: import warnings
warnings.filterwarnings('ignore')

import torch
import torch.optim as optim
from sklearn.metrics import classification_report

from src import (
    LipSequenceDataset, SequenceAugmentation, Conv3DNet, ImprovedConv3DNet,
    validate, train_model, GradCAM3D, BalancedFocalLoss,
    setup_device, calculate_class_weights, create_data_loaders, print_summary,
    plot_training_curves, plot_confusion_matrices, visualize_gradcam,
    visualize_model_architecture,
    DATA_DIR, TRAIN_SUBJECTS, VAL_SUBJECTS, TEST_SUBJECTS, WARMUP_EPOCHS,
    WINDOW_SIZES, BATCH_SIZE, NUM_EPOCHS, LEARNING_RATE, NUM_CLASSES,
    CLASS_NAMES,
    WEIGHT_DECAY, EARLY_STOPPING_PATIENCE, MAX_GRAD_NORM, FOCAL_LOSS_GAMMA
)

print(" All imports successful")
```

All imports successful

```
[2]: device = setup_device(seed=42)
print(f" Using device: {device}")

# Display GPU information if available
if torch.cuda.is_available():
    print(f" GPU: {torch.cuda.get_device_name(0)}")
    print(f" CUDA Version: {torch.version.cuda}")
    print(f" GPU Memory: {torch.cuda.get_device_properties(0).total_memory /
    1024**3:.2f} GB")
else:
    print(" CUDA not available, using CPU (training will be slower)")
```

Using device: cuda
GPU: NVIDIA GeForce RTX 3090
CUDA Version: 12.4
GPU Memory: 24.00 GB

0.1 2. Model Architecture

```
[3]: sample_model = Conv3DNet(num_classes=NUM_CLASSES, window_size=9)
      visualize_model_architecture(sample_model, input_shape=(1, 3, 9, 100, 180))
```

Model Architecture:

```
=====
Conv3DNet(
  (conv1): Sequential(
    (0): Conv3d(3, 64, kernel_size=(3, 3, 3), stride=(1, 1, 1), padding=(1, 1, 1))
    (1): BatchNorm3d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): Dropout3d(p=0.15, inplace=False)
    (4): MaxPool3d(kernel_size=(1, 2, 2), stride=(1, 2, 2), padding=0,
dilation=1, ceil_mode=False)
  )
  (conv2): Sequential(
    (0): Conv3d(64, 128, kernel_size=(3, 3, 3), stride=(1, 1, 1), padding=(1, 1, 1))
    (1): BatchNorm3d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): Dropout3d(p=0.25, inplace=False)
    (4): MaxPool3d(kernel_size=(2, 2, 2), stride=(2, 2, 2), padding=0,
dilation=1, ceil_mode=False)
  )
  (conv3): Sequential(
    (0): Conv3d(128, 256, kernel_size=(3, 3, 3), stride=(1, 1, 1), padding=(1, 1, 1))
    (1): BatchNorm3d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): Dropout3d(p=0.35, inplace=False)
    (4): MaxPool3d(kernel_size=(1, 2, 2), stride=(1, 2, 2), padding=0,
dilation=1, ceil_mode=False)
  )
  (conv4): Sequential(
    (0): Conv3d(256, 512, kernel_size=(3, 3, 3), stride=(1, 1, 1), padding=(1, 1, 1))
    (1): BatchNorm3d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): Dropout3d(p=0.45, inplace=False)
    (4): MaxPool3d(kernel_size=(1, 2, 2), stride=(1, 2, 2), padding=0,
dilation=1, ceil_mode=False)
  )
)
```

```

(temporal_attention): Sequential(
  (0): Conv3d(512, 256, kernel_size=(3, 1, 1), stride=(1, 1, 1), padding=(1, 0, 0))
  (1): ReLU(inplace=True)
  (2): Conv3d(256, 512, kernel_size=(3, 1, 1), stride=(1, 1, 1), padding=(1, 0, 0))
  (3): Sigmoid()
)
(avgpool): AdaptiveAvgPool3d(output_size=(1, 1, 1))
(fc1): Sequential(
  (0): Dropout(p=0.5, inplace=False)
  (1): Linear(in_features=512, out_features=256, bias=True)
  (2): ReLU(inplace=True)
  (3): BatchNorm1d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
)
(fc2): Sequential(
  (0): Dropout(p=0.5, inplace=False)
  (1): Linear(in_features=256, out_features=128, bias=True)
  (2): ReLU(inplace=True)
)
(classifier): Linear(in_features=128, out_features=6, bias=True)
)
=====

```

Total parameters: 5,605,638

Trainable parameters: 5,605,638

Input shape: (1, 3, 9, 100, 180)

Output shape: torch.Size([1, 6])

0.2 3. Training

Train models for different window sizes (5, 7, 9).

```

[4]: results = {}

for window_size in WINDOW_SIZES:
    print(f"\n{'='*60}")
    print(f"Training with window_size = {window_size}")
    print(f"{'='*60}\n")

    # Create datasets
    train_dataset = LipSequenceDataset(DATA_DIR, TRAIN_SUBJECTS,
    ↪window_size=window_size, transform=SequenceAugmentation(p=0.5), mode='train')
    val_dataset = LipSequenceDataset(DATA_DIR, VAL_SUBJECTS,
    ↪window_size=window_size, transform=None, mode='val')

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```

    test_dataset = LipSequenceDataset(DATA_DIR, TEST_SUBJECTS,
    ↪window_size=window_size, transform=None, mode='test')

    # Calculate class weights and create data loaders
    class_weights, class_weights_tensor = calculate_class_weights(train_dataset.
    ↪labels, NUM_CLASSES)
    print(f"Class weights: {class_weights}")

    train_loader, val_loader, test_loader = create_data_loaders(train_dataset,
    ↪val_dataset, test_dataset, class_weights, BATCH_SIZE)

    # Initialize model
    model = ImprovedConv3DNet(num_classes=NUM_CLASSES, window_size=window_size).
    ↪to(device)
    criterion = BalancedFocalLoss(alpha=class_weights_tensor.to(device),
    ↪gamma=FOCAL_LOSS_GAMMA)
    optimizer = optim.Adam(model.parameters(), lr=LEARNING_RATE,
    ↪weight_decay=WEIGHT_DECAY)
    scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer, mode='min',
    ↪factor=0.5, patience=5)

    # Train model
    history = train_model(model, train_loader, val_loader, criterion,
    ↪optimizer, scheduler, NUM_EPOCHS, device, patience=EARLY_STOPPING_PATIENCE,
    ↪max_grad_norm=MAX_GRAD_NORM, warmup_epochs=WARMUP_EPOCHS)

    # Evaluate on test set
    model.load_state_dict(history['best_model_state'])
    test_loss, test_acc, test_preds, test_labels = validate(model, test_loader,
    ↪criterion, device)

    print(f'\nTest Results for window_size={window_size}:')
    print(f'Test Loss: {test_loss:.4f}, Test Acc: {test_acc:.2f}%')
    print('\nClassification Report:')
    print(classification_report(test_labels, test_preds,
    ↪target_names=CLASS_NAMES))

    results[window_size] = {**history, 'test_acc': test_acc, 'test_loss':
    ↪test_loss, 'test_preds': test_preds, 'test_labels': test_labels, 'model':
    ↪model}
    print(f"\n Completed training for window_size={window_size}\n")

```

```

=====
Training with window_size = 5
=====

```

Created 3456 sequences for train set (window_size=5)
Class distribution: Counter({0: 1516, 4: 484, 3: 463, 2: 425, 5: 313, 1: 255})
Created 573 sequences for val set (window_size=5)
Class distribution: Counter({0: 246, 1: 69, 3: 67, 5: 65, 2: 64, 4: 62})
Created 1060 sequences for test set (window_size=5)
Class distribution: Counter({0: 548, 2: 124, 1: 110, 3: 100, 5: 92, 4: 86})
Class weights: {0: 0.37994722955145116, 1: 2.2588235294117647, 2:
1.3552941176470588, 3: 1.244060475161987, 4: 1.1900826446280992, 5:
1.840255591054313}

Epoch 1/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.98it/s,
loss=1.7405, acc=18.75%]

Train Loss: 1.9452, Train Acc: 18.75%

Val Loss: 1.3850, Val Acc: 15.36%

New best validation accuracy: 15.36%

Epoch 2/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.01it/s,
loss=1.5913, acc=24.10%]

Train Loss: 1.8065, Train Acc: 24.10%

Val Loss: 1.2705, Val Acc: 19.72%

New best validation accuracy: 19.72%

Epoch 3/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.97it/s,
loss=1.4654, acc=26.79%]

Train Loss: 1.6770, Train Acc: 26.79%

Val Loss: 1.1624, Val Acc: 22.34%

New best validation accuracy: 22.34%

Epoch 4/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.08it/s,
loss=1.5519, acc=30.56%]

Train Loss: 1.5381, Train Acc: 30.56%

Val Loss: 1.0140, Val Acc: 25.13%

New best validation accuracy: 25.13%

Epoch 5/60 [Training]: 100%| | 108/108 [00:27<00:00, 4.00it/s,
loss=1.2953, acc=32.32%]

Train Loss: 1.4753, Train Acc: 32.32%

Val Loss: 0.8943, Val Acc: 30.19%

New best validation accuracy: 30.19%

Epoch 6/60 [Training]: 100%| | 108/108 [00:28<00:00, 3.73it/s,
loss=0.9531, acc=34.64%]

Train Loss: 1.3473, Train Acc: 34.64%

Val Loss: 0.8474, Val Acc: 30.54%

New best validation accuracy: 30.54%

Epoch 7/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.94it/s,
loss=1.7024, acc=37.27%]

Train Loss: 1.2733, Train Acc: 37.27%
 Val Loss: 0.8607, Val Acc: 27.75%
 Epoch 8/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.91it/s,
 loss=1.2317, acc=37.99%]
 Train Loss: 1.2026, Train Acc: 37.99%
 Val Loss: 0.8175, Val Acc: 32.46%
 New best validation accuracy: 32.46%
 Epoch 9/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.90it/s,
 loss=1.2733, acc=41.03%]
 Train Loss: 1.1590, Train Acc: 41.03%
 Val Loss: 0.6863, Val Acc: 34.55%
 New best validation accuracy: 34.55%
 Epoch 10/60 [Training]: 100%| | 108/108 [00:29<00:00, 3.72it/s,
 loss=1.2790, acc=41.12%]
 Train Loss: 1.1312, Train Acc: 41.12%
 Val Loss: 0.8322, Val Acc: 30.19%
 Epoch 11/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.96it/s,
 loss=0.7424, acc=42.39%]
 Train Loss: 1.1453, Train Acc: 42.39%
 Val Loss: 0.7568, Val Acc: 30.37%
 Epoch 12/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.99it/s,
 loss=1.3256, acc=43.23%]
 Train Loss: 1.0908, Train Acc: 43.23%
 Val Loss: 0.7274, Val Acc: 32.29%
 Epoch 13/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.93it/s,
 loss=0.6764, acc=42.62%]
 Train Loss: 1.0993, Train Acc: 42.62%
 Val Loss: 0.7645, Val Acc: 30.89%
 Epoch 14/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.07it/s,
 loss=1.0745, acc=42.68%]
 Train Loss: 1.0557, Train Acc: 42.68%
 Val Loss: 0.6940, Val Acc: 33.68%
 Epoch 15/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.99it/s,
 loss=1.0930, acc=43.75%]
 Train Loss: 1.0503, Train Acc: 43.75%
 Val Loss: 0.6346, Val Acc: 34.21%
 Epoch 16/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.06it/s,
 loss=1.1277, acc=43.84%]

Train Loss: 1.0216, Train Acc: 43.84%
Val Loss: 0.7764, Val Acc: 31.41%

Epoch 17/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.13it/s,
loss=1.2039, acc=45.37%]

Train Loss: 1.0205, Train Acc: 45.37%
Val Loss: 0.7431, Val Acc: 33.33%

Epoch 18/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.05it/s,
loss=0.7718, acc=46.15%]

Train Loss: 0.9777, Train Acc: 46.15%
Val Loss: 0.7313, Val Acc: 35.95%
New best validation accuracy: 35.95%

Epoch 19/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.11it/s,
loss=1.0437, acc=45.37%]

Train Loss: 0.9832, Train Acc: 45.37%
Val Loss: 0.6318, Val Acc: 35.25%

Epoch 20/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.12it/s,
loss=0.9810, acc=44.30%]

Train Loss: 0.9582, Train Acc: 44.30%
Val Loss: 0.7851, Val Acc: 28.45%

Epoch 21/60 [Training]: 100%| | 108/108 [00:28<00:00, 3.85it/s,
loss=0.6597, acc=47.28%]

Train Loss: 0.9617, Train Acc: 47.28%
Val Loss: 0.6579, Val Acc: 37.35%
New best validation accuracy: 37.35%

Epoch 22/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.94it/s,
loss=0.9699, acc=45.40%]

Train Loss: 0.9410, Train Acc: 45.40%
Val Loss: 0.6388, Val Acc: 38.05%
New best validation accuracy: 38.05%

Epoch 23/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.89it/s,
loss=0.7378, acc=47.97%]

Train Loss: 0.9428, Train Acc: 47.97%
Val Loss: 0.6223, Val Acc: 38.39%
New best validation accuracy: 38.39%

Epoch 24/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.96it/s,
loss=0.9161, acc=46.56%]

Train Loss: 0.8952, Train Acc: 46.56%
Val Loss: 0.6682, Val Acc: 34.73%

Epoch 25/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.90it/s,
loss=0.7702, acc=47.45%]

Train Loss: 0.8978, Train Acc: 47.45%
Val Loss: 0.6247, Val Acc: 36.13%

Epoch 26/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.90it/s,
loss=0.8386, acc=48.09%]

Train Loss: 0.8773, Train Acc: 48.09%
Val Loss: 0.5803, Val Acc: 39.09%
New best validation accuracy: 39.09%

Epoch 27/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.94it/s,
loss=0.9859, acc=50.03%]

Train Loss: 0.8457, Train Acc: 50.03%
Val Loss: 0.6415, Val Acc: 33.16%

Epoch 28/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.92it/s,
loss=0.8381, acc=50.00%]

Train Loss: 0.8400, Train Acc: 50.00%
Val Loss: 0.5514, Val Acc: 37.87%

Epoch 29/60 [Training]: 100%| | 108/108 [00:28<00:00, 3.75it/s,
loss=0.9166, acc=50.61%]

Train Loss: 0.8568, Train Acc: 50.61%
Val Loss: 0.5504, Val Acc: 38.05%

Epoch 30/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.89it/s,
loss=1.1887, acc=51.33%]

Train Loss: 0.8412, Train Acc: 51.33%
Val Loss: 0.6075, Val Acc: 39.27%
New best validation accuracy: 39.27%

Epoch 31/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.02it/s,
loss=0.9184, acc=50.78%]

Train Loss: 0.7906, Train Acc: 50.78%
Val Loss: 0.6761, Val Acc: 37.17%

Epoch 32/60 [Training]: 100%| | 108/108 [00:28<00:00, 3.80it/s,
loss=0.7237, acc=52.78%]

Train Loss: 0.7919, Train Acc: 52.78%
Val Loss: 0.6242, Val Acc: 38.57%

Epoch 33/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.88it/s,
loss=0.6986, acc=52.46%]

Train Loss: 0.7956, Train Acc: 52.46%
Val Loss: 0.5894, Val Acc: 40.14%
New best validation accuracy: 40.14%

Epoch 34/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.03it/s,
loss=1.0452, acc=51.68%]

Train Loss: 0.7958, Train Acc: 51.68%
Val Loss: 0.6334, Val Acc: 32.11%

Epoch 35/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.03it/s,
loss=1.4566, acc=51.50%]

Train Loss: 0.7868, Train Acc: 51.50%
Val Loss: 0.6143, Val Acc: 38.22%

Epoch 36/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.04it/s,
loss=0.6846, acc=54.28%]

Train Loss: 0.7280, Train Acc: 54.28%
Val Loss: 0.5787, Val Acc: 38.74%

Epoch 37/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.98it/s,
loss=0.5731, acc=56.66%]

Train Loss: 0.6734, Train Acc: 56.66%
Val Loss: 0.5750, Val Acc: 38.74%

Epoch 38/60 [Training]: 100%| | 108/108 [00:28<00:00, 3.86it/s,
loss=1.0465, acc=55.61%]

Train Loss: 0.6806, Train Acc: 55.61%
Val Loss: 0.5388, Val Acc: 40.66%

New best validation accuracy: 40.66%

Epoch 39/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.86it/s,
loss=1.0252, acc=55.73%]

Train Loss: 0.6623, Train Acc: 55.73%
Val Loss: 0.5541, Val Acc: 40.14%

Epoch 40/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.98it/s,
loss=0.5250, acc=57.12%]

Train Loss: 0.6772, Train Acc: 57.12%
Val Loss: 0.5403, Val Acc: 39.97%

Epoch 41/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.04it/s,
loss=0.6104, acc=57.00%]

Train Loss: 0.6804, Train Acc: 57.00%
Val Loss: 0.5273, Val Acc: 41.36%

New best validation accuracy: 41.36%

Epoch 42/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.04it/s,
loss=0.7098, acc=57.87%]

Train Loss: 0.6447, Train Acc: 57.87%
Val Loss: 0.5213, Val Acc: 40.66%

Epoch 43/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.03it/s,
loss=0.6975, acc=58.19%]
Train Loss: 0.6246, Train Acc: 58.19%
Val Loss: 0.5309, Val Acc: 41.01%

Epoch 44/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.01it/s,
loss=0.3931, acc=60.16%]
Train Loss: 0.6271, Train Acc: 60.16%
Val Loss: 0.5330, Val Acc: 40.84%

Epoch 45/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.05it/s,
loss=0.4490, acc=59.81%]
Train Loss: 0.6159, Train Acc: 59.81%
Val Loss: 0.5413, Val Acc: 39.97%

Epoch 46/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.05it/s,
loss=0.7750, acc=60.47%]
Train Loss: 0.5948, Train Acc: 60.47%
Val Loss: 0.5460, Val Acc: 38.39%

Epoch 47/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.03it/s,
loss=0.5650, acc=61.05%]
Train Loss: 0.5848, Train Acc: 61.05%
Val Loss: 0.5564, Val Acc: 39.97%

Epoch 48/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.92it/s,
loss=0.6271, acc=59.81%]
Train Loss: 0.6034, Train Acc: 59.81%
Val Loss: 0.6106, Val Acc: 37.52%

Epoch 49/60 [Training]: 100%| | 108/108 [00:29<00:00, 3.65it/s,
loss=0.4949, acc=61.31%]
Train Loss: 0.5873, Train Acc: 61.31%
Val Loss: 0.5149, Val Acc: 41.36%

Epoch 50/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.99it/s,
loss=0.5198, acc=61.40%]
Train Loss: 0.5872, Train Acc: 61.40%
Val Loss: 0.5476, Val Acc: 40.14%

Epoch 51/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.05it/s,
loss=0.5204, acc=62.30%]
Train Loss: 0.5431, Train Acc: 62.30%
Val Loss: 0.5251, Val Acc: 41.54%
New best validation accuracy: 41.54%

Epoch 52/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.04it/s,
loss=0.4876, acc=63.05%]

Train Loss: 0.5251, Train Acc: 63.05%
 Val Loss: 0.5455, Val Acc: 38.74%
 Epoch 53/60 [Training]: 100%| | 108/108 [00:28<00:00, 3.83it/s,
 loss=0.7277, acc=62.70%]
 Train Loss: 0.5314, Train Acc: 62.70%
 Val Loss: 0.5557, Val Acc: 41.19%
 Epoch 54/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.93it/s,
 loss=0.6788, acc=63.60%]
 Train Loss: 0.5273, Train Acc: 63.60%
 Val Loss: 0.5395, Val Acc: 41.54%
 Epoch 55/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.97it/s,
 loss=0.5302, acc=63.83%]
 Train Loss: 0.4934, Train Acc: 63.83%
 Val Loss: 0.5193, Val Acc: 41.71%
 New best validation accuracy: 41.71%
 Epoch 56/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.98it/s,
 loss=0.2847, acc=64.29%]
 Train Loss: 0.4831, Train Acc: 64.29%
 Val Loss: 0.5522, Val Acc: 40.66%
 Epoch 57/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.94it/s,
 loss=0.7993, acc=65.42%]
 Train Loss: 0.5291, Train Acc: 65.42%
 Val Loss: 0.5246, Val Acc: 41.54%
 Epoch 58/60 [Training]: 100%| | 108/108 [00:26<00:00, 4.01it/s,
 loss=0.4990, acc=62.91%]
 Train Loss: 0.4972, Train Acc: 62.91%
 Val Loss: 0.5473, Val Acc: 41.88%
 New best validation accuracy: 41.88%
 Epoch 59/60 [Training]: 100%| | 108/108 [00:27<00:00, 3.99it/s,
 loss=0.3327, acc=64.99%]
 Train Loss: 0.5006, Train Acc: 64.99%
 Val Loss: 0.5802, Val Acc: 39.79%
 Epoch 60/60 [Training]: 100%| | 108/108 [00:28<00:00, 3.86it/s,
 loss=1.1100, acc=63.22%]
 Train Loss: 0.5114, Train Acc: 63.22%
 Val Loss: 0.5611, Val Acc: 42.58%
 New best validation accuracy: 42.58%
 Test Results for window_size=5:
 Test Loss: 0.7364, Test Acc: 43.58%

Classification Report:

	precision	recall	f1-score	support
Silence/Neutral	0.92	0.22	0.35	548
/A/	0.63	0.76	0.69	110
/I/	0.28	0.73	0.41	124
/U/	0.42	0.52	0.46	100
/E/	0.65	0.57	0.61	86
/O/	0.25	0.73	0.37	92
accuracy			0.44	1060
macro avg	0.52	0.59	0.48	1060
weighted avg	0.68	0.44	0.43	1060

Completed training for window_size=5

=====
Training with window_size = 7
=====

Created 2298 sequences for train set (window_size=7)
Class distribution: Counter({0: 1023, 4: 320, 3: 306, 2: 278, 5: 207, 1: 164})
Created 381 sequences for val set (window_size=7)
Class distribution: Counter({0: 162, 1: 48, 3: 47, 4: 43, 5: 41, 2: 40})
Created 704 sequences for test set (window_size=7)
Class distribution: Counter({0: 371, 2: 81, 1: 67, 3: 66, 5: 63, 4: 56})
Class weights: {0: 0.37438905180840665, 1: 2.3353658536585367, 2:
1.3776978417266188, 3: 1.2516339869281046, 4: 1.196875, 5: 1.8502415458937198}

Epoch 1/60 [Training]: 100%| | 72/72 [00:26<00:00, 2.69it/s,
loss=1.6025, acc=16.54%]

Train Loss: 1.9903, Train Acc: 16.54%
Val Loss: 1.4779, Val Acc: 12.34%
New best validation accuracy: 12.34%

Epoch 2/60 [Training]: 100%| | 72/72 [00:26<00:00, 2.72it/s,
loss=2.0907, acc=21.98%]

Train Loss: 1.9252, Train Acc: 21.98%
Val Loss: 1.4127, Val Acc: 18.11%
New best validation accuracy: 18.11%

Epoch 3/60 [Training]: 100%| | 72/72 [00:26<00:00, 2.67it/s,
loss=1.6477, acc=24.67%]

Train Loss: 1.8636, Train Acc: 24.67%
Val Loss: 1.3493, Val Acc: 19.16%

New best validation accuracy: 19.16%

Epoch 4/60 [Training]: 100%| | 72/72 [00:27<00:00, 2.64it/s, loss=1.7535, acc=23.72%]

Train Loss: 1.7503, Train Acc: 23.72%
Val Loss: 1.2234, Val Acc: 21.52%
New best validation accuracy: 21.52%

Epoch 5/60 [Training]: 100%| | 72/72 [00:27<00:00, 2.63it/s, loss=1.6698, acc=27.42%]

Train Loss: 1.6647, Train Acc: 27.42%
Val Loss: 1.1320, Val Acc: 24.15%
New best validation accuracy: 24.15%

Epoch 6/60 [Training]: 100%| | 72/72 [00:27<00:00, 2.59it/s, loss=1.6796, acc=28.11%]

Train Loss: 1.6345, Train Acc: 28.11%
Val Loss: 1.1377, Val Acc: 28.35%
New best validation accuracy: 28.35%

Epoch 7/60 [Training]: 100%| | 72/72 [00:26<00:00, 2.70it/s, loss=1.7462, acc=29.50%]

Train Loss: 1.5926, Train Acc: 29.50%
Val Loss: 1.0783, Val Acc: 29.13%
New best validation accuracy: 29.13%

Epoch 8/60 [Training]: 100%| | 72/72 [00:26<00:00, 2.69it/s, loss=1.5179, acc=32.77%]

Train Loss: 1.4706, Train Acc: 32.77%
Val Loss: 0.9166, Val Acc: 29.66%
New best validation accuracy: 29.66%

Epoch 9/60 [Training]: 100%| | 72/72 [00:27<00:00, 2.67it/s, loss=1.5237, acc=35.25%]

Train Loss: 1.4089, Train Acc: 35.25%
Val Loss: 0.9317, Val Acc: 27.56%

Epoch 10/60 [Training]: 100%| | 72/72 [00:27<00:00, 2.66it/s, loss=1.8838, acc=33.94%]

Train Loss: 1.3889, Train Acc: 33.94%
Val Loss: 0.9035, Val Acc: 28.08%

Epoch 11/60 [Training]: 100%| | 72/72 [00:28<00:00, 2.53it/s, loss=1.3912, acc=36.81%]

Train Loss: 1.3466, Train Acc: 36.81%
Val Loss: 0.7785, Val Acc: 30.71%
New best validation accuracy: 30.71%

Epoch 12/60 [Training]: 100%| | 72/72 [00:26<00:00, 2.73it/s,
loss=1.1638, acc=38.90%]
Train Loss: 1.3046, Train Acc: 38.90%
Val Loss: 0.8327, Val Acc: 26.25%

Epoch 13/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.81it/s,
loss=0.9371, acc=38.38%]
Train Loss: 1.2295, Train Acc: 38.38%
Val Loss: 0.7438, Val Acc: 32.28%
New best validation accuracy: 32.28%

Epoch 14/60 [Training]: 100%| | 72/72 [00:26<00:00, 2.75it/s,
loss=0.7832, acc=42.65%]
Train Loss: 1.1727, Train Acc: 42.65%
Val Loss: 1.0148, Val Acc: 27.82%

Epoch 15/60 [Training]: 100%| | 72/72 [00:27<00:00, 2.64it/s,
loss=1.2290, acc=39.95%]
Train Loss: 1.1954, Train Acc: 39.95%
Val Loss: 0.7689, Val Acc: 29.92%

Epoch 16/60 [Training]: 100%| | 72/72 [1:16:41<00:00, 63.92s/it,
loss=1.1131, acc=42.52%]
Train Loss: 1.1654, Train Acc: 42.52%
Val Loss: 0.8369, Val Acc: 31.50%

Epoch 17/60 [Training]: 100%| | 72/72 [00:26<00:00, 2.69it/s,
loss=0.7565, acc=41.08%]
Train Loss: 1.1127, Train Acc: 41.08%
Val Loss: 0.9410, Val Acc: 30.71%

Epoch 18/60 [Training]: 100%| | 72/72 [00:26<00:00, 2.71it/s,
loss=0.9399, acc=41.78%]
Train Loss: 1.1539, Train Acc: 41.78%
Val Loss: 0.8660, Val Acc: 29.66%

Epoch 19/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.79it/s,
loss=2.1659, acc=42.30%]
Train Loss: 1.1071, Train Acc: 42.30%
Val Loss: 0.6961, Val Acc: 34.91%
New best validation accuracy: 34.91%

Epoch 20/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.79it/s,
loss=1.4854, acc=43.52%]
Train Loss: 1.0473, Train Acc: 43.52%
Val Loss: 0.7924, Val Acc: 33.07%

Epoch 21/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.80it/s,
loss=0.9044, acc=43.86%]

Train Loss: 1.0595, Train Acc: 43.86%

Val Loss: 0.6483, Val Acc: 35.17%

New best validation accuracy: 35.17%

Epoch 22/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.78it/s,
loss=1.2200, acc=42.73%]

Train Loss: 1.0324, Train Acc: 42.73%

Val Loss: 1.0084, Val Acc: 30.18%

Epoch 23/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.81it/s,
loss=0.9231, acc=44.73%]

Train Loss: 0.9952, Train Acc: 44.73%

Val Loss: 0.9165, Val Acc: 30.71%

Epoch 24/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.78it/s,
loss=1.1849, acc=43.65%]

Train Loss: 1.0686, Train Acc: 43.65%

Val Loss: 0.6719, Val Acc: 37.01%

New best validation accuracy: 37.01%

Epoch 25/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.82it/s,
loss=1.0334, acc=46.91%]

Train Loss: 0.9517, Train Acc: 46.91%

Val Loss: 0.7594, Val Acc: 33.60%

Epoch 26/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.77it/s,
loss=0.9473, acc=44.87%]

Train Loss: 0.9875, Train Acc: 44.87%

Val Loss: 0.9089, Val Acc: 32.28%

Epoch 27/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.80it/s,
loss=1.2571, acc=45.87%]

Train Loss: 0.9733, Train Acc: 45.87%

Val Loss: 0.6668, Val Acc: 35.96%

Epoch 28/60 [Training]: 100%| | 72/72 [00:26<00:00, 2.75it/s,
loss=0.8876, acc=48.74%]

Train Loss: 0.9594, Train Acc: 48.74%

Val Loss: 0.9142, Val Acc: 33.33%

Epoch 29/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.80it/s,
loss=0.6853, acc=46.91%]

Train Loss: 0.9391, Train Acc: 46.91%

Val Loss: 0.8699, Val Acc: 31.50%

Epoch 30/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.79it/s,
loss=0.8102, acc=48.09%]

Train Loss: 0.8939, Train Acc: 48.09%
Val Loss: 0.6646, Val Acc: 35.43%

Epoch 31/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.81it/s,
loss=0.5262, acc=50.13%]

Train Loss: 0.8741, Train Acc: 50.13%
Val Loss: 0.8155, Val Acc: 35.43%

Epoch 32/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.79it/s,
loss=1.4055, acc=51.09%]

Train Loss: 0.8626, Train Acc: 51.09%
Val Loss: 0.8298, Val Acc: 30.45%

Epoch 33/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.80it/s,
loss=0.7330, acc=50.09%]

Train Loss: 0.8970, Train Acc: 50.09%
Val Loss: 0.6638, Val Acc: 34.38%

Epoch 34/60 [Training]: 100%| | 72/72 [00:25<00:00, 2.78it/s,
loss=0.7070, acc=49.70%]

Train Loss: 0.8466, Train Acc: 49.70%
Val Loss: 0.7288, Val Acc: 35.70%
Early stopping at epoch 34

Test Results for window_size=7:
Test Loss: 0.7445, Test Acc: 53.12%

Classification Report:

	precision	recall	f1-score	support
Silence/Neutral	0.87	0.47	0.61	371
/A/	0.58	0.73	0.64	67
/I/	0.22	0.44	0.29	81
/U/	0.34	0.76	0.47	66
/E/	0.54	0.34	0.42	56
/O/	0.65	0.75	0.70	63
accuracy			0.53	704
macro avg	0.53	0.58	0.52	704
weighted avg	0.67	0.53	0.55	704

Completed training for window_size=7

```
=====
Training with window_size = 9
=====
```

```
Created 1719 sequences for train set (window_size=9)
Class distribution: Counter({0: 764, 3: 240, 4: 236, 2: 215, 5: 150, 1: 114})
Created 285 sequences for val set (window_size=9)
Class distribution: Counter({0: 128, 1: 36, 5: 31, 3: 31, 2: 30, 4: 29})
Created 527 sequences for test set (window_size=9)
Class distribution: Counter({0: 273, 2: 57, 1: 52, 5: 51, 3: 50, 4: 44})
Class weights: {0: 0.375, 1: 2.513157894736842, 2: 1.3325581395348838, 3:
1.19375, 4: 1.2139830508474576, 5: 1.91}
```

```
Epoch 1/60 [Training]: 100%|      | 54/54 [00:25<00:00,  2.16it/s,
loss=1.6171, acc=16.99%]
```

```
Train Loss: 2.0254, Train Acc: 16.99%
```

```
Val Loss: 1.4325, Val Acc: 12.63%
```

```
  New best validation accuracy: 12.63%
```

```
Epoch 2/60 [Training]: 100%|      | 54/54 [00:24<00:00,  2.18it/s,
loss=1.7832, acc=19.43%]
```

```
Train Loss: 1.9902, Train Acc: 19.43%
```

```
Val Loss: 1.4576, Val Acc: 11.93%
```

```
Epoch 3/60 [Training]: 100%|      | 54/54 [00:24<00:00,  2.18it/s,
loss=1.6925, acc=20.07%]
```

```
Train Loss: 1.9541, Train Acc: 20.07%
```

```
Val Loss: 1.4478, Val Acc: 25.61%
```

```
  New best validation accuracy: 25.61%
```

```
Epoch 4/60 [Training]: 100%|      | 54/54 [00:24<00:00,  2.16it/s,
loss=2.0017, acc=21.18%]
```

```
Train Loss: 1.9599, Train Acc: 21.18%
```

```
Val Loss: 1.3593, Val Acc: 21.40%
```

```
Epoch 5/60 [Training]: 100%|      | 54/54 [00:24<00:00,  2.17it/s,
loss=1.7733, acc=24.84%]
```

```
Train Loss: 1.8250, Train Acc: 24.84%
```

```
Val Loss: 1.2845, Val Acc: 20.00%
```

```
Epoch 6/60 [Training]: 100%|      | 54/54 [00:25<00:00,  2.16it/s,
loss=1.5845, acc=25.89%]
```

```
Train Loss: 1.7770, Train Acc: 25.89%
```

```
Val Loss: 1.2963, Val Acc: 17.54%
```

```
Epoch 7/60 [Training]: 100%|      | 54/54 [00:24<00:00,  2.16it/s,
loss=1.5274, acc=26.99%]
```

Train Loss: 1.7310, Train Acc: 26.99%
 Val Loss: 1.1861, Val Acc: 22.46%
 Epoch 8/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.10it/s,
 loss=1.7578, acc=23.73%]
 Train Loss: 1.7633, Train Acc: 23.73%
 Val Loss: 1.1390, Val Acc: 23.86%
 Epoch 9/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.12it/s,
 loss=1.6134, acc=28.91%]
 Train Loss: 1.6784, Train Acc: 28.91%
 Val Loss: 1.1143, Val Acc: 24.56%
 Epoch 10/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.08it/s,
 loss=1.6153, acc=30.19%]
 Train Loss: 1.6642, Train Acc: 30.19%
 Val Loss: 1.1921, Val Acc: 23.86%
 Epoch 11/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.10it/s,
 loss=1.4816, acc=30.31%]
 Train Loss: 1.5556, Train Acc: 30.31%
 Val Loss: 1.0444, Val Acc: 28.42%
 New best validation accuracy: 28.42%
 Epoch 12/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.09it/s,
 loss=1.5785, acc=31.30%]
 Train Loss: 1.5010, Train Acc: 31.30%
 Val Loss: 1.0807, Val Acc: 23.51%
 Epoch 13/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.12it/s,
 loss=1.5137, acc=33.04%]
 Train Loss: 1.4682, Train Acc: 33.04%
 Val Loss: 0.9774, Val Acc: 28.07%
 Epoch 14/60 [Training]: 100%| | 54/54 [00:26<00:00, 2.01it/s,
 loss=1.7424, acc=35.78%]
 Train Loss: 1.4404, Train Acc: 35.78%
 Val Loss: 1.0259, Val Acc: 25.26%
 Epoch 15/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.11it/s,
 loss=1.3647, acc=36.53%]
 Train Loss: 1.4393, Train Acc: 36.53%
 Val Loss: 0.9440, Val Acc: 24.91%
 Epoch 16/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.12it/s,
 loss=1.2680, acc=38.98%]

Train Loss: 1.3545, Train Acc: 38.98%
 Val Loss: 0.9884, Val Acc: 28.77%
 New best validation accuracy: 28.77%

Epoch 17/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.12it/s,
 loss=1.1339, acc=36.65%]

Train Loss: 1.3788, Train Acc: 36.65%
 Val Loss: 0.9193, Val Acc: 27.37%

Epoch 18/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.12it/s,
 loss=1.6695, acc=37.23%]

Train Loss: 1.3905, Train Acc: 37.23%
 Val Loss: 0.7957, Val Acc: 32.63%
 New best validation accuracy: 32.63%

Epoch 19/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.11it/s,
 loss=1.5916, acc=38.22%]

Train Loss: 1.3171, Train Acc: 38.22%
 Val Loss: 0.8230, Val Acc: 30.18%

Epoch 20/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.12it/s,
 loss=1.5871, acc=38.34%]

Train Loss: 1.2803, Train Acc: 38.34%
 Val Loss: 0.7737, Val Acc: 30.53%

Epoch 21/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.12it/s,
 loss=1.2058, acc=39.44%]

Train Loss: 1.2142, Train Acc: 39.44%
 Val Loss: 0.8963, Val Acc: 30.18%

Epoch 22/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.10it/s,
 loss=1.5047, acc=38.98%]

Train Loss: 1.2951, Train Acc: 38.98%
 Val Loss: 0.7987, Val Acc: 30.88%

Epoch 23/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.11it/s,
 loss=1.2523, acc=38.39%]

Train Loss: 1.2628, Train Acc: 38.39%
 Val Loss: 0.8835, Val Acc: 31.23%

Epoch 24/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.10it/s,
 loss=1.0068, acc=41.77%]

Train Loss: 1.2135, Train Acc: 41.77%
 Val Loss: 0.7884, Val Acc: 27.02%

Epoch 25/60 [Training]: 100%| | 54/54 [00:25<00:00, 2.10it/s,
 loss=1.5514, acc=38.63%]

```

Train Loss: 1.2036, Train Acc: 38.63%
Val Loss: 0.8804, Val Acc: 29.82%

Epoch 26/60 [Training]: 100%|      | 54/54 [00:25<00:00,  2.11it/s,
loss=0.9739, acc=42.35%]

Train Loss: 1.1465, Train Acc: 42.35%
Val Loss: 0.9063, Val Acc: 30.88%

Epoch 27/60 [Training]: 100%|      | 54/54 [00:25<00:00,  2.12it/s,
loss=1.4651, acc=42.82%]

Train Loss: 1.0943, Train Acc: 42.82%
Val Loss: 0.7949, Val Acc: 31.58%

Epoch 28/60 [Training]: 100%|      | 54/54 [00:25<00:00,  2.10it/s,
loss=0.8891, acc=41.71%]

Train Loss: 1.1486, Train Acc: 41.71%
Val Loss: 0.8917, Val Acc: 28.77%
Early stopping at epoch 28

Test Results for window_size=9:
Test Loss: 1.0544, Test Acc: 20.11%

```

Classification Report:

	precision	recall	f1-score	support
Silence/Neutral	1.00	0.01	0.01	273
/A/	0.47	0.71	0.56	52
/I/	0.31	0.26	0.28	57
/U/	0.12	0.22	0.15	50
/E/	0.55	0.14	0.22	44
/O/	0.12	0.69	0.20	51
accuracy			0.20	527
macro avg	0.43	0.34	0.24	527
weighted avg	0.67	0.20	0.15	527

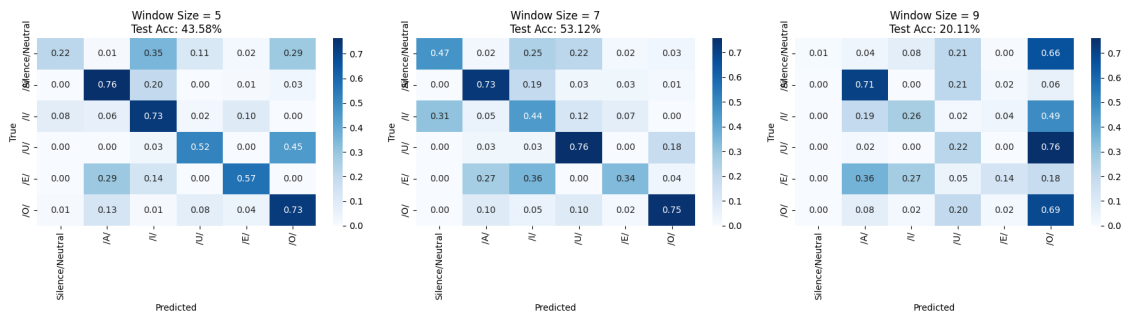
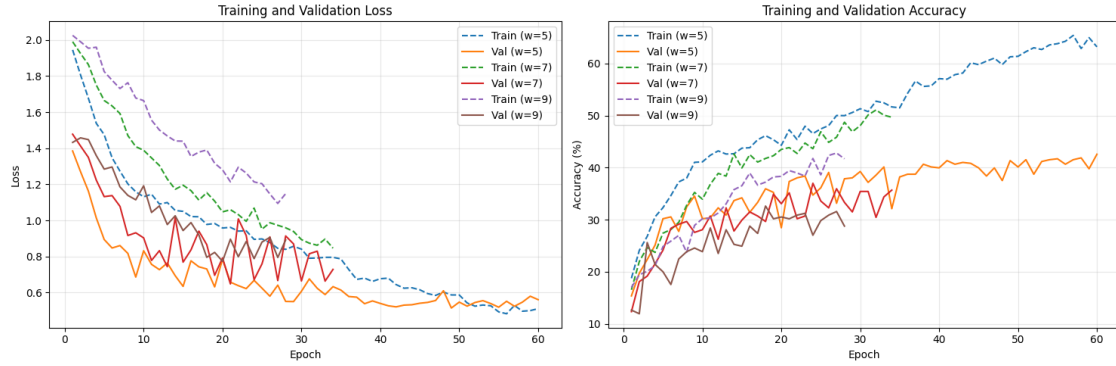
Completed training for window_size=9

0.3 4. Results Visualization

```

[5]: plot_training_curves(results, WINDOW_SIZES)
     plot_confusion_matrices(results, WINDOW_SIZES, CLASS_NAMES)
     print_summary(results, WINDOW_SIZES, CLASS_NAMES)

```



SUMMARY OF RESULTS

Window Size	Val Acc (%)	Test Acc (%)	Test Loss
5	42.58	43.58	0.7364
7	37.01	53.12	0.7445
9	32.63	20.11	1.0544

Best model: window_size=7 with test accuracy=53.12%

0.4 5. Grad-CAM Visualization

```
[6]: if results:
    best_window_size = max(results.keys(), key=lambda k: results[k]['test_acc'])
    print(f"Visualizing with best model (window_size={best_window_size},
    ↪test_acc={results[best_window_size]['test_acc']:.2f}%)")

    model = results[best_window_size]['model']
```

```

# Select appropriate layer for GradCAM based on model type
if hasattr(model, 'conv4'):
    target_layer = model.conv4[0] # Conv3DNet
elif hasattr(model, 'res_block3'):
    target_layer = model.res_block3[0] # ImprovedConv3DNet
else:
    # Fallback: find last Conv3d layer
    conv_layers = [m for m in model.modules() if isinstance(m, torch.nn.
↳Conv3d)]
    if conv_layers:
        target_layer = conv_layers[-1]
    else:
        raise ValueError("No Conv3d layer found in model for GradCAM")

gradcam = GradCAM3D(model, target_layer)

test_dataset = LipSequenceDataset(DATA_DIR, TEST_SUBJECTS,
↳window_size=best_window_size, transform=None, mode='test')
if len(test_dataset) > 0:
    sample_sequence, sample_label = test_dataset[0]
    sample_sequence = sample_sequence.unsqueeze(0).to(device)
    visualize_gradcam(model, gradcam, sample_sequence, sample_label,
↳best_window_size, CLASS_NAMES)
else:
    print(" No test samples available.")
else:
    print(" No results available. Please run training first.")

```

Visualizing with best model (window_size=7, test_acc=53.12%)

Created 704 sequences for test set (window_size=7)

Class distribution: Counter({0: 371, 2: 81, 1: 67, 3: 66, 5: 63, 4: 56})

